MILITARY SPECIFICATION

SAFETY AND ARMING DEVICE, GUIDED MISSILE: M143 (M143E1) LOGIC CIRCUIT HYBRID FOR

This Specification is approved for use within the U. S. Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance, test, manufacture and acceptance requirements for hybrid microcircuits, hereinafter to be called hybrids, to be used in the M143 (M143E1) Safety and Arming (S&A) Device.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-S-766 - Steel Plates, Sheets, and Strip-Corrosion Resisting

MILITARY

MIL-I-23011 - Iron-Nickel Alloys for Sealing to Glasses and Ceramics
MIL-C-26074 - Coating, Electroless Nickel, Requirements for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-M-38510 - Microcircuits, General Specifications for
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specifications for
MIL-S-48687 - Safety and Arming Device, Guided Missile: M143E1 Loading Assembling and Packing for
MIL-M-55565 - Microcircuits, Preparation for Delivery of
MIL-S-63257 - Safety and Arming Device, Guided Missile: M143 Loading Assembling and Packing for

STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts
MIL-STD-883 - Test Methods and Procedures for Microelectronics

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9311370 - Logic Circuit Hybrid

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the procuring activitiy or as directed by the Contracting Officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials.

3.1.1 Parts, materials and processes. Parts, materials and processes shall be in accordance with the applicable drawings, specifications and standards.

3.1.2 Metals. External metal surfaces shall be corrosion resistant or shall be plated or treated to resist corrosion.

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3.1.3 Other materials. External parts, elements or coatings including markings shall be inherently non-nutrient to fungus and shall not blister, crack, outgas, soften flow or exhibit defects that adversely affect storage, operation or environmental capabilities of hybrids.

3.1.4 Case material and finish. The case material shall conform to one of the following:

a. Iron, Nickel, Cobalt alloy (KOVAR) per MIL-I-23011, Class 1, with a Rockwell B hardness of 70 typical and a grain size of 6 ± 1. The KOVAR shall be plated with 150 ± 50 microinches of electroless nickel per MIL-C-26074, Class 1.

b. Steel, Corrosion Resisting (Stainless steel) per Specification QQ-S-766, Class 304.

3.1.5 Lead material and finish. The lead material shall consist of Iron, Nickel, Cobalt alloy (KOVAR) per MIL-I-23011, Class 1, with a Rockwell B hardness of 70 typical and a grain size of 6 ± 1. The external lead finish shall conform to one of the following:

a. Hot solder dip. The hot solder dip shall be homogeneous with a minimum thickness at the crest of the major flats of 200 microinches of solder (SN60 or SN63) over the primary finishes in accordance with (b) or (c) below or over nickel plate with a plating thickness of 100 microinches minimum and 200 microinches maximum.

b. Bright acid tin plate. Thickness of 100 microinches minimum and 400 microinches maximum.

c. Gold plate. Gold plating shall be a mimimum of 99.7 percent gold (0.3 percent maximum for all impurities and other metals combined). Gold plating shall be a minimum of 50 microinches and a maximum of 225 microinches thick.

d. Other finish. Any finish which meets the solderability requirements of MIL-STD-883, Method 2003 and has been approved by the procuring activity.

3.1.6 Internal lead finish. The finish for the portion of the lead internal to the hybrid shall be compatible with the type of wire and the bonding method used.

3.2 Design and construction. Hybrid design and construction shall be in accordance with the requirements specified herein and in the applicable drawings.

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3.2.1 Package. All hybrids shall be hermetically sealed in glass, metal or ceramic (or combinations of these) packages. No organic or polymeric materials (lacquers, varnishes, coatings, adhesives, greases, etc.) shall be used inside the hybrid package. Adhesives may be used only for attachment of components to substrate and substrate to case.

3.2.2 Design documentation. When specified in the procurement document, the topography for all hybrids shall be submitted to the procuring activity. Design documentation shall be traceable to the specific part, drawing or type numbers to which it applies, and to the production lot(s) and inspection lot codes under which hybrids are manufactured and tested so that revisions can be identified.

3.2.3 Internal conductors. Internal conductors (metallization stripes, contact areas, bonding interfaces, wires or other current carrying elements) of the hybrids shall be designed so that no properly fabricated conductor shall experience, in normal operation, a current density in excess of 2 x 10E5 amperes/centimeter squared, including allowances for normal production tolerances on design dimensions, and actual thicknesses at critical areas such as steps in elevation or contact windows.

3.2.4 Pre-seal bake. The hybrids shall be baked at 200°C in a dry nitrogen atmosphere for twenty-four (24) hours minimum immediately prior to sealing.

3.2.4.1 Alternative. The hybrids shall be baked at 125°C in a vacuum with a maximum absolute pressure of ten(10) inches of mercury (Hg) for two hours immediately prior to sealing.

3.2.5 Stabilization bake. Capped hybrids, after being subjected to Stabilization Bake in accordance with MIL-STD-883, Method 1008, Test Condition C for 48 hours, shall meet the final electrical requirements herein specified.

3.3 Logic circuit hybrid (dwg. 9311370). All measurements required in this specification shall be taken with a measurement device that has an input impedance equal to or greater than 10 megaohms and an open circuit potential no greater than 10.0VDC. Prior to start and for duration of testing the following load and biasing resistors shall be added externally to the hybrid under test.

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PIN (+) to (-)	RESISTOR VALUE (OHMS)
12 - 29	100K *
13 - 29	100K
14 - 29	1K
21 - 29	10K
24 - 29	1K (2 watt min.)
26 - 29	47K
27 - 29	47K
28 - 29	10K
30 - 29	47K
41 - 29	1K

* K = Kilohms = 1,000

Unless otherwise specified, all resistors are 1/4 watt min., ± 10%.

3.3.1 Continuity and resistance. The electrical resistance shall be as specified in Table I.

TABLE I. Continuity and resistance

PIN (+) to (-)	REQUIREMENTS (OHMS)	PIN (+) to (-)	REQUIREMENTS (OHMS)
		37 to 29	88K to 132K
		36 to 29	88K to 132K
24 to 25	1.0 MAX.	36 to 37	120K to 240K
29 to Case	1.0 MAX.	37 to 36	120K to 240K
20 to 19	90 to 110	20 to 29	0.9 to 1.1 MEG *
1 to 29	90 to 110	22 to 29	0.9 to 1.1 MEG
44 to 29	90 to 110	23 to 29	0.9 to 1.1 MEG
16 to 15	Less than 19 MEG.	18 to 17	0.99 to 1.01 MEG
33 to 29	80K to 120K	42 to 43	0.99 to 1.01 MEG
35 to 36	120K to 240K	15 to 16	Greater than 20 MEG
35 to 37	120K to 240K		

* MEG = Megaohms = 1,000,000

3.3.2 Charge command test.

3.3.2.1 TM monitors. With 100.0 + 5.0 VDC applied to Converter input Pins (+) 13 (right) and (-) 29 and followed by 31.0 - 0.5 VDC applied to Charge Command input Pins (+) 27 and (-) 29, the voltage across the Pins to (-) 29 shall be specified in Table II.

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TABLE II. TM outputs

IDENTIFICATION	PIN (+)	REQUIREMENT (volts)
a. POWER/DESTRUCT	26	-0.5 to +0.5
b. 5VDC TEST	30	-0.5 to +0.5
c. ITL TM	34	3.5 to 5.5
d. FIRE TM	32	-0.5 to +0.5
e. ARM TM	38	3.5 to 5.5
f. ROTOR MONITOR, Right	36	3.5 to 5.5
g. ROTOR MONITOR, Left	37	3.5 to 5.5
h. 5 VDC (REGULATOR) MONITOR	40	3.5 to 5.5
i. POWER output (DIODE DROP)	24	28.5 to 30.0
j. ITL MONITOR Input	35	3.5 to 5.5
k. CONVERTER INPUT, Left	12	-0.5 to +0.5
l. 5 VDC CAPACITOR	21	3.5 to 5.5
m. 5 VDC CAPACITOR	28	3.5 to 5.5

3.3.2.2 Fire TM. With the voltage applied as specified in 3.3.2.1, the Fire TM voltage at Pin 32 shall be 4.5 $\pm$ 1.0 VDC when 5.0 $\pm$ 0.5 VDC is applied to Pin 33.

3.3.2.3 Arm TM. With the voltage applied as specified in 3.3.2.1, the Arm TM voltage at Pin 38 shall be 0 $\pm$ 0.5 VDC, when Pin 36 is connected to Pin 29. With Pin 36 removed from Pin 29 and Pin 37 connected to Pin 29 the ARM TM shall be 0 $\pm$ 0.5 VDC.

3.3.2.4 ITL TM. With the voltage applied as specified in 3.3.2.1, the ITL TM voltage at Pin 34 shall be 0 $\pm$ 0.5 VDC when a 100 ohm $\pm$ 10 percent, 1/4 watt min., resistor is connected between Pin 35 and Pin 29 (ground).

3.3.2.5 Convertor TM. With the Charge Command voltage applied and 0.5 seconds min. after the right convertor voltage is removed as specified in 3.3.2.1, the convertor TM at Pin 39 shall be 0 $\pm$ 0.5 VDC and the destruct Pin 14 (right) and Pin 41 (left) shall be 1.5 VDC min.

3.3.2.6 Destruct pulse.

3.3.2.6.1 Power up. After the voltage is applied as specified in 3.5.2.1, the destruct pulse at Pin 14 right and Pin 41 left shall change from 0 $\pm$ 0.5 VDC min. to 1.5 VDC min. in less than 50 milliseconds. The duration of the destruct pulse shall be not less than 2 milliseconds and not greater than 200 milliseconds.

3.3.2.6.2 Converter input removed. With the voltage applied as specified in 3.3.2.1, for 0.5 seconds min., the destruct pulse at Pin (+) 14 (right) and Pin (+) 41 (left) shall change from 0 $\pm$ 0.5 VDC to 1.5 VDC min. in not less than 10 milliseconds and not greater than 50 milliseconds after the right converter voltage is removed.

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3.3.3 Power/destruct tests.

3.3.3.1 TM monitors. With 100.0 + 5.0 VDC applied to convertor pins (+)12 (left) and (-)29 followed by 23.0 + 0.5 VDC applied to power/destruct input pins (+)26 and (-)29, the voltage across the pins to (-)29 shall be as specified in Table III.

TABLE III. TM outputs

	IDENTIFICATION	PIN (+)	REQUIREMENT (volts)
a.	CHARGE COMMAND	27	-0.5 to +0.5
b.	5VDC TEST	30	-0.5 to +0.5
c.	ITL TM	34	3.5 to 5.5
d.	FIRE TM	32	0.5 to +0.5
e.	ARM TM	38	3.5 to 5.5
f.	ROTOR MONITOR, Right	36	3.5 to 5.5
g.	ROTOR MONITOR, Left	37	3.5 to 5.5
h.	5 VDC (REGULATOR) MONITOR	40	3.5 to 5.5
i.	POWER OUTPUT	24	21.0 to 22.5
j.	CONVERTOR INPUT, Right	13	-0.5 to +0.5
k.	ITL MONITOR INPUT	35	3.5 to 5.5
l.	5 VDC CAPACITOR	21	3.5 to 5.5
m.	5 VDC CAPACITOR	28	3.5 to 5.5

3.3.3.2 Power TM threshold. With the voltage applied as specified in 3.3.3.1, the Power TM voltage at Pin 31 shall be 4.5 $\pm$ 1.0 VDC, when the Power/Destruct input applied to Pin 26 is equal to or greater than 25.0 VDC. The Power TM voltage at Pin 31 shall be 0 $\pm$ 0.5 VDC, when the Power/Destruct input applied to Pin 26 is less than 21.0 VDC. This transition shall occur at 23.0 $\pm$ 2.0VDC.

3.3.3.3 Destruct pulse threshold. With the voltage applied as specified in 3.3.3.1, the Destruct voltage at both Pins 14 and 41 shall be 0 $\pm$ 0.5 VDC when the Power/Destruct input applied to Pin 26 is equal to or greater than 20.5 VDC. The Destruct voltage at both Pins 14 and 41 shall be 1.5 VDC min., when the Power/Destruct input applied to Pin 26 is less than 18.5 VDC. This transition shall occur at 19.5 $\pm$ 1.0VDC.

3.3.3.4 Convertor threshold. With the voltage applied as specified in 3.3.3.1, the destruct voltage at Pin 14 and Pin 41 shall be 0 $\pm$ 0.5 VDC and the convertor TM at Pin 39 shall be 4.5 $\pm$ 1.0 VDC when the voltage on Pin 12 is greater than 70.0 VDC. The destruct voltage at Pins 14 and 41 shall be 1.5 VDC min. and the converter TM Pin 39 shall be 0 $\pm$ 0.5 VDC when the voltage on Pin 12 is less than 60.0 VDC. This transition shall occur at 65.0 $\pm$ 5.0 VDC.

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3.3.4 Five volt test. Apply 5.0 + 0.5 VDC from pin (+)21 to pin (-)29 and pin (+)28 to pin (-)29, 100 + 5 VDC to left convertor pins (+)12 and (-)29, 23.0 + 0.5 VDC to logic board power output pins (+)24 and (-)29 and 5.0 ± 0.5 VDC to 5 VDC test pins (+)30 and (-)29 for the following tests.

3.3.4.1 Destruct pulse. With the voltage applied as specified in 3.3.4, wait a minimum of 0.5 seconds, then remove the 5 VDC at Pin 30 specified in 3.3.4. The destruct pulse at Pin (+)14 right and Pin (+)41 left shall change from 0 ± 0.5 VDC to 1.5 VDC min. in not less than 10 milliseconds and not greater than 50 milliseconds.

3.3.4.2 Five volt threshold. With the voltage applied as specified in 3.3.4, the Destruct pulse at both Pin (+)14 right and Pin (+)41 left shall be 0 ± 0.5 VDC when the voltage at Pin (+)30 is set to 3.5 + 2.0 VDC. The destruct pulse at both Pin (+)14 right and Pin (+)41 left shall be 1.5 VDC min. when the voltage at Pin (+)30 is less than 1.5 VDC. This transition shall occur at 2.5 ± 1.0 VDC.

3.4 Environmental and mechanical.

3.4.1 High temperature storage. Hybrids, after being subjected to high temperature storage per MIL-STD-883, Method 1008, 150°C temperature for 1000 hours duration, shall meet the final electrical requirements herein specified.

3.4.2 Temperature cycling. Hybrids, after being subjected to temperature cycling as specified in MIL-STD-883, Method 1010, Test Condition C and cycling between steps 1 and 3, shall meet the final electrical requirements herein specified.

3.4.3 Operating temperature. The hybrids shall meet the final electrical requirements specified herein at any temperature from -40°F -5°F (-40°C -2.8°C) to 150°F +5°F (66°C +2.8°C). Unless otherwise indicated, the temperature shall be at ambient 77°F ± 5.4°F (25°C ± 3°C) temperature.

3.4.4 Seal.

3.4.4.1 Fluorocarbon gross leak. Hybrids shall meet the requirements specified in MIL-STD-883, Method 1014, Test Condition C2.

3.4.4.2 Tracer gas (HE) fine leak. Hybrids shall meet the requirements specified in MIL-STD-883, Method 1014, Test Condition A1.

3.4.5 Thermal shock. Hybrids, after being subjected to thermal shock as specified in MIL-STD-883, Method 1011, Test Condition B, 15 cycles, shall meet the final electrical requirements after exposure.

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3.4.6 Mechanical shock. Hybrids, after being subjected to mechanical shock as specified in MIL-STD-883, Method 2002, Test Condition B, shall meet the final electrical requirements after exposure.

3.4.7 Vibration, variable frequency. Hybrids, after being subjected to vibration as specified in MIL-STD-883, Method 2007, Test Condition A, shall meet the final electrical requirements.

3.4.8 Lead integrity. Hybrids shall meet the requirements of MIL-STD-883, Method 2004, Test Condition B1.

3.4.9 Bond strength. Hybrid wirebonds, when tested as specified in MIL-STD-883, Method 2011, Test Condition D, shall exhibit a minimum bond strength as follows:

WIRE MATERIAL	WIRE DIAMETER	MINIMUM BOND STRENGTH
Au	1 mil *	3 grams
Au	2 mil	6 grams
Al	1.25 mil	2 grams
Al	2 mil	3.8 grams
Al	3 mil	8 grams
Al	4 mil	14 grams
Al	5 mil	22 grams
Al	8 mil	54 grams
Al	10 mil	80 grams

* Mil = 10^{-3}

3.4.10 Burn-in. Hybrids after being subjected to burn-in in accordance with the detailed drawing and MIL-STD-883, Method 1015, Test Condition B, for 168 hours at +125°C minimum, shall meet the final electrical requirements herein specified.

3.4.11 Internal visual. (Pre Seal) Hybrids shall meet the requirements of MIL-STD-883, Method 2017.

3.4.12 External visual. Hybrids shall meet the requirements of MIL-STD-883, Method 2009.

3.4.13 Constant acceleration. Hybrids, when tested in accordance with MIL-STD-883, Method 2001, Test Condition B, Y1 axis, shall meet the final electrical requirements herein specified.

3.4.14 End point electrical. Hybrids shall meet the requirements herein specified.

3.4.15 Final electrical. Hybrids shall meet the electrical requirements herein specified.

3.4.16 Internal visual (post stress). Hybrids shall meet the requirements of 4.5.3.

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3.4.17 Solderability. Hybrids shall meet the solerability requirements specified in MIL-STD-883, Method 2003.

3.5 Marking. Marking shall be in accordance with the detailed drawing and shall include as a minimum the following:

a. Index point
b. Procuring activity part number
c. Lot number and date (month and year)
d. Serial number
e. Code identification number

3.5.1 Marking location. The Lot Number and Date, Serial Number and procuring activity part number shall be marked on the frontside as a minimum. The code identification number may be marked on the front or backside. The index point shall be marked as specified in the detailed drawing.

3.5.2 Marking permanency and legibility. Marking shall be correct, permanent, legible and capable of withstanding the requirements of MIL-STD-202, Method 215.

3.6 Rework provisions. All rework permitted on hybrids shall be accomplished in accordance with procedures and safeguards of MIL-M-38510 and available for review by the procuring activity. Rework of the package seal for hybrids shall be permitted a maximum of two times after final seal and only for packages which have an outer seal perimeter of at least three inches in total length. Allowable rework of sealed hybrids shall be limited to recleaning of any hybrid or portion thereof, restamping to correct defective marking, and lead straightening (provided the reworked hybrids meet the requirements of "lead integrity").

3.6.1 In-process rework.

3.6.1.1 Substrate rework.

a. One scratched, open or discontinuous metallization path or conductor pattern may be repaired by bridging with a minimum of two bonding wires having equivalent current carrying capacity.

b. Metallization paths containing voids may be repaired as in a. above or, where approved by the procuring activity, by an epoxy void-repair procedure.

3.6.1.2 Rebonding and element replacement.

a. The total number of wire rebond attempts shall be limited to a maximum of 10 percent of the total number (or the nearest whole number) of bonds in the hybrids. A bond shall be defined as a wire to post or wire to pad bond (i.e., for a 14 lead wire-bonded package there are 28 bonds). Bond-offs required to

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clear the bonder after an unsuccessful first bond attempt need not be considered as rebonds provided they can be identified as bond-offs by being made physically off the plated post or if they contain a non-typical number of bonding tool marks. The initial bond attempt need not be visible. The replacement of one wire bonded at one end shall be considered as one rebond; a replacement of a wire bonded at both ends, or an unsuccessful bond attempt of a wire already bonded at the other end shall be considered as two rebonds. Compounding bonding is not permissible.

b. A maximum of two replacements of each element or die shall be permitted, except that there shall be no more than 20% replacement of the total elements or dice. Within the 20% replacement allowance, replacement of active devices following electrical testing, as defined in detailed drawing, shall be limited to 10% of total elements or dice.

c. No more than one rebond attempt shall be permitted at any pad, and no rebonds shall be made directly over an area where metallization of intended wire bond areas has been lifted.

d. A hybrid may be subjected to a maximum total of four interconnect bonding or element replacement cycles following completion of the first electrical test as defined by the detailed drawing (i.e. after completion of the first electrical test cycle, the hybrid cannot be subjected to more than four rework/repair cycles). Rebonding to substrate pad or package post when performed to effect element replacement shall not be counted against the total rebond limitations. Element removal will not be performed using techniques which expose other dice to temperatures in excess of 150°C.

3.6.2 Delidding and rework of sealed hybrids. Delidding and rework of sealed hybrids shall be permitted with the following limitations:

a. Delidding, rework and reseal of a hybrid shall be permitted a maximum of one time following initial package seal.

b. Rework of a sealed hybrid may be accomplished at any point in the normal manufacturing process flow prior to the Quality Conformance Inspection lot acceptance testing of 4.4.2.2.

c. All resealed hybrids shall be subjected to the requirements of 4.4.2.1. All resealed hybrids shall be formed into unique lots and shall be subjected to the requirements of 4.4.2.2.

d. The total cumulative in-process and post-seal rework shall not exceed that specified in 3.6.1.

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3.7 First article. When specified in the contract of purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.8 Product assurance program. The manufacturer shall establish a product assurance program in accordance with the requirements of MIL-M-38510, Appendix A. After receipt of First Article acceptance, the manufacturer shall not implement any changes in the product assurance program documentation. When the process control or quality control documents listed in the approved product assurance program plan are changed, the manufacturer shall notify the procuring activity.

3.9 Test data and samples. Test data required in 4.4.2.1, 4.4.2.2 and 4.4.2.3 shall be submitted if requested by the procuring activity, as specified in Appendix A of MIL-S-63257 or MIL-S-48687 as applicable. Samples of nonstandard parts may be required by the procuring activity. These samples shall be submitted in quantities and to the destination specified by the procuring activity for tests and examination. Sample quantities to be specified will not exceed six units of any part. Sample parts and materials may be tested to destruction by the procuring activity and will not be returned to the contractor. When there is more than one supplier for a part, parts from each supplier shall be considered for seperate submission.

3.10 Workmanship. The component shall be fabricated and finished in a thorough, workmanlike manner. It shall be free of burrs, chips, sharp edges, unblended radii, cracks, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning methods used shall not be injurious to any part, nor shall parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. See MIL-A-48078. The following types of inspection shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection.

4.3.1 Sample. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

PART NUMBER	DESCRIPTION	QUANTITY
9311370	Logic Circuit Hybrid	65

4.3.1.1 Homogeneity. The first article shall comply with the homogeneity criteria of MIL-A-48078.

4.3.2 Inspection to be performed. The first article sample may be subjected by the Government to any or all of the requirements of the applicable drawings and to the inspections of Table I. No defects are permitted, except for bond strength and solderability as defined by the LTPD (Lot Tolerance Percent Defective). All defects are classified as major unless otherwise specified.

4.3.3 Rejection. If any assembly, component, or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of any assembly, component or test specimen to comply with any of the requirements.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.3.4	Logic Circuit Hybrid		See 4.3.1
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Group I - 1/ 5/	15			
	Physical Dimensions			3.	Dwg 9311370/Gage
	Marking Permanence			3.5.2	MIL-STD-202
	External Visual			3.4.12	MIL-STD-883
	Solderability		2/	3.6.17	MIL-STD-883
	Lead Integrity			3.4.8	MIL-STD-883
	Seal			3.4.4	MIL-STD-883
	Bond Strength		3/	3.4.9	MIL-STD-883
	Group II - 1/	15			
	Thermal Shock			3.4.5	MIL-STD-883
	Temperature Cycling			3.4.2	MIL-STD-883
	Seal			3.4.4	MIL-STD-883
	External Visual			3.4.12	MIL-STD-883
	Final Electrical			3.4.15	4.4.2.2/Test

NOTES: 1/ Each unit in the group shall be subjected to all the examinations or tests in that group unless otherwise specified.

2/ An LTPD=15 (Lot Tolerance Percent Defective) applied to hybrid leads per MIL-M-38510 Table B-1, shall be used for ACC/REJ limits. Minimum sample size shall be 2 hybrids.

3/ An LTPD=7 shall be applied to the wirebonds, with minimum sample size of 2 hybrids. Equal number of wirebonds shall be tested of each, initially.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER See 4.3.1
4.3.4	Logic Circuit Hybrid (continued)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Group III - 1/	15			
	Mechanical Shock			3.4.6	MIL-STD-883
	Vibration, Variable Frequency			3.4.7	MIL-STD-883
	Constant Acceleration			3.4.13	MIL-STD-883
	Seal			3.4.4	MIL-STD-883
	Final Electrical			3.4.15	4.4.2.2/Test
	Internal Visual			3.4.16	4.5.3/Visual
	Group IV - 1/	20			
	End Point Electrical 4/			3.4.14	4.4.2.2/Test
	High Temperature Storage			3.4.1	MIL-STD-883
	Final Electrical			3.4.15	4.4.2.2/Test

NOTES: 4/ 50% of units shall be tested at high temperature; (150°F +5) and 50% shall be tested at low temperature; (-40°F -5°).

5/ Electrical Reject units may be utilized for tests of Group I, however Reject units must have been subjected to the test environments.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot function provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

a. 100 percent test. If any unit under test fails to comply with any of the applicable requirements the unit shall be rejected.

b. Sample test. If any test specimen fails to comply with any of the applicable requirements, the lot shall be rejected. Lots shall not be resubmitted for these tests without prior approval of the procuring activity. Acceptance of product shall be discontinued pending investigation of the failure and approval by the design activity of corrective action proposed by the contractor.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311370
4.4.2.1	Logic Circuit Hybrid (Screening Test)				NEXT HIGHER ASSEMBLY 9311372
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101	Internal Visual (Pre Seal)		100%	3.4.11	MIL-STD-883
102	Preseal Bake		100%	3.2.4	Test
103	Stabilization Bake		100%	3.2.5	MIL-STD-883
104	Temperature Cycling		100%	3.4.2	MIL-STD-883
105	Constant Acceleration		100%	3.4.13	MIL-STD-883
106	Seal		100%	3.4.4	MIL-STD-883
107	Final Electrical		100%	3.3	4.4.2.2
108	Burn-IN		100%	3.4.10	MIL-STD-883
109	Final Electrical		100%	3.3	4.4.2.2
110	External Visual		100%	3.4.12	MIL-STD-883
111	Electrical Tests at -40°F to -45°F		0.65%	3.3	4.4.2.2
112	Electrical Tests at +150°F to 155°F		0.65%	3.3	4.4.2.2
113	Case Material & Finish		.....	3.1.4	Certificate of Conf.
114	Lead Material & Finish		.....	3.1.5	Certificate of Conf.

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63256C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311370
4.4.2.2	Logic Circuit Hybrid (Electrical Test)				NEXT HIGHER ASSEMBLY 9311372
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Continuity and resistance		100%	3.3.1	Test, Fluke 8000A
102	TM monitors (with charge command and right convertor input)		100%	3.3.2.1	Test
103	Fire TM		100%	3.3.2.2	
104	ARM TM		100%	3.3.2.3	
105	ITL TM		100%	3.3.2.4	
106	Convertor TM		100%	3.3.2.5	
107	Destruct pulse (Power Up)		100%	3.3.2.6.1	Test
108	Destruct pulse (Convertor Input Removed)		100%	3.3.2.6.2	Test
109	TM monitors (with Power/Destruct and Left convertor input)		100%	3.3.3.1	Test
110	Power TM threshold (with Power/Destruct input)		100%	3.3.3.2	Test
111	Destruct pulse Threshold (with Power/ Destruct input)		100%	3.3.3.3	Test
112	Convertor TM threshold (with Power/ Destruct input)		100%	3.3.3.4	Test
113	Five Volt Tests (Destruct pulse)		100%	3.3.4.1	Test
114	Five Volt Tests (Five Volt Threshold)		100%	3.3.4.2	Test

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63256C (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Logic Circuit Hybrid (Lot Acceptance Test)	1 OF 2	9311370
			NEXT HIGHER ASSEMBLY 9311372

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Groups I Through III Inspection Lot Size — Sample Size 1-250 — 2 251-575 — 4				
Major					
101	Group I (Non Destructive) - Note 1 Physical Dimensions	100%	Note: (3)	3.0	Dwg 9311370/Gage
102	Group II (Destructive) - Note 1 Visual - Internal - Note 2	100%	(3)	3.4.11	MIL-STD-883
103	Group III (Destructive) Solderability - 25% of the Terminations	50%	(3)	3.4.17	MIL-STD-883
104	Bond Strength - 14 of the Bonds shall be pulled for minimum strength, 14 additional Bonds shall be pulled to destruction for information on each unit.	100%	(3)	3.4.9	MIL-STD-883

NOTES:
1. Group I samples may be used in Group IV or V. Electrical rejects may be used for Group II.
2. Requires removal of Cover (Hybrids may be cleaned).
3. Accept on 0 defects, reject on 1 or more defects.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Logic Circuit Hybrid (Lot Acceptance Test) (Continued)	2 OF 2	9311370
			NEXT HIGHER ASSEMBLY 9311372

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Groups IV (Note 2)				
	Inspection Lot Size — Sample Size: 1-100 — 3; 101-250 — 5; 251-475 — 7				
	Group IV (Destructive) (Note 1)		Note:		
105	Thermal Shock	100%	(3)	3.4.5	MIL-STD-883
106	Temperature Cycling	100%	(3)	3.4.2	MIL-STD-883
107	Seal	100%	(3)	3.4.4	MIL-STD-883
108	Electrical Tests	100%	(3)	3.4.15	Test/4.4.2.2
	Group V - (Note 2)				
	Inspection Lot Size — Sample Size: 1-100 — 3; 101-250 — 5; 251-475 — 7				
	Group V (Destructive) (Note 1)				
109	Mechanical Shock	100%	(3)	3.4.6	MIL-STD-883
110	Vibration Variable Frequency	100%	(3)	3.4.7	MIL-STD-883
111	Seal	100%	(3)	3.4.4	MIL-STD-883
112	Electrical Tests	100%	(3)	3.4.15	Test/4.4.2.2

NOTES:
1. Each unit in group shall be subjected to all the examinations or tests in that group.
2. Groups IV & V require separate samples, group samples shall not be intermixed.
3. Accept on 0 defects, reject on 1 or more defects.

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4.4.3 Rejection. If any unit fails the tests of 4.4.2.1, the unit shall be rejected. If any test specimen fails the tests of 4.4.2.2 the lot shall be rejected unless otherwise specified. Lots shall not be resubmitted for these tests without prior approval of the procuring activity. If directed by the procuring activity, acceptance of additional hybrids of the same type shall be discontinued pending investigation of the failure and approval by the procuring activity of corrective action proposed by the contractor. Upon approval from the procuring activity, rejected lots may be reworked per 3.5.2 and resubmitted for these tests.

4.4.4 Testing. All tests are specified in 4.4.2.1 and 4.4.2.2.

4.4.5 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed in this specification is identified, either directly or by reference, in the "Paragraph/Reference/Inspection Method" column of the First Article or Quality Conformance Inspection Tables herein. The contractor shall submit inspection equipment designs to the Government for approval in accordance with the terms of the contract. See 6.3.

4.5 Methods of inspection.

4.5.1 Wire bond strength. Wire bonds shall be pulled to determine the minimum strength. The minimum number of wires pulled in each sample hybrid shall be (for each wire gage): One (1) wire from each type* of transistor, diode, capacitor, and resistor die; three (3) wires from each type of integrated circuit; and five (5) wires from a header to substrate. In no case shall the total number of wires pulled per sample hybrid be less than 15. Any failures shall be described, their location identified, and the stress value at which failure occurred shall be recorded. After the above testing, the same bonds shall be tested to destruction, for information.

*"Type" shall be defined as any attribute which would result in the component being different, such as:

a. Physical height of the bonding surface above the substrate.

b. Manufacturer.

c. Bond pad material composition.

d. Number of bonds per device.

e. Die topography

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4.5.2 Stabilization bake. Sealed hybrids shall be subjected to stabilization bake in accordance with MIL-STD-883, Method 1008, Test Condition C.

4.5.3 Internal visual. Following stress testing, hybrids shall be delidded and internal visual examination performed at 30X magnification. No evidence of damage or dislocation is permitted.

4.5.4 Test tolerances. All tolerances specified herein are absolute with no allowance made for measuring equipment inaccuracy. Measurements shall ensure that the values measured are within the tolerances or limits specified.

5. PACKAGING

5.1 Packaging. Hybrids shall be protected individually to prevent damage from electrostatic field forces in accordance with the requirements of MIL-M-55565. Other field force protection may be employed as necessary to prevent damage to the hybrids. Preservation-packaging, packing and container marking shall be in accordance with level C requirements of MIL-M-55565 unless otherwise specified in the procurement document.

6. NOTES

6.1 Intended use. The hybrids covered by this specification are intended for use on the Safety and Arming device: M143 (M143E1).

6.2 Ordering data. See MIL-A-48078.

6.2.1 Data requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
3.9	Test and Demonstration Reports	DI-T-1906

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

6.2.2 Submission of data. Submit data, as required, to Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAR-R (D), Dover, New Jersey 07801-5001.

6.3 Submission of designs for approval. See MIL-A-48078. Submit equipment designs, as required, to U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAR-I (D), Dover, New Jersey 07801-5001.

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6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as, US Army Armament Research and Development Command (ARRADCOM) or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.5 Key word listing.

Safe & Arming Device M143
Hybrid

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1336-A533)

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